

Problem-Solution Fit

Date	6 th July 2026
Team ID	SWTID-2026-4078
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Banks and financial institutions • Credit analysts and loan officers • Customers applying for credit cards	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited financial information • Incomplete application details • Varying income sources • Limited credit history • Dependence on manual processing	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual Verification: Accurate but slow and costly • Traditional Rule-Based Systems: Fast but lacks adaptability • Machine Learning Models: Fast, scalable, accurate and learns from historical data
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual verification is slow and time-consuming • Human errors during evaluation • Large number of applications received daily • Delays and inconsistent approval decisions	9. PROBLEM ROOT / CAUSE [RC] • Increasing application volume • Manual verification process • Inconsistent human decision-making • Incomplete applicant data • Inefficient risk assessment	7. BEHAVIOR + ITS INTENSITY [BE] • Applicants expect quick approval decisions • Banks process thousands of applications daily • Need for high-speed, consistent and reliable evaluation
3. TRIGGERS TO ACT [TR] • Increase in credit card applications • Need for faster approval • Customer demand for instant decisions • Requirement to reduce operational costs	10. YOUR SOLUTION [SL] Develop a Credit Card Approval Prediction System using Machine Learning (Logistic Regression, Decision Tree, Random Forest and XGBoost) integrated with a Flask web application.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Bank websites • Mobile banking applications • Online credit card portals
4. EMOTIONS BEFORE / AFTER [EM] Before: Frustration, uncertainty, long waiting time and stress After: Faster approval, confidence, transparency, improved customer satisfaction and trust	The system predicts approval or rejection instantly based on applicant information, improving accuracy, reducing processing time and assisting banks in making reliable decisions.	OFFLINE • Bank branches • Customer service centers • Financial consultants